

Lecture 1

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This lecture will give a brief introduction to the theory of abelian varieties, largely following van der Geer's book [1].

1 Abelian varieties

Let k be a field.

Definition 1.1. An abelian variety A/k is a proper smooth connected group scheme over k

Remark 1.2. [2, Section 0BF9] In fact, there are a lot equivalent definitions for abelian variety A/k . Let

- projective, proper,
- geometrically irreducible, irreducible, geometrically connected, connected,
- smooth, geometrically reduced

be three sets of properties, pick one from each of them, and let A be a group scheme over k with the chosen properties. Then A is an abelian variety.

Definition 1.3. Let X be a k -group scheme and $x \in X(k)$. We define the right translation $t_x : X \rightarrow X$ to be the composition

$$X \cong X \times \operatorname{Spec} k \xrightarrow{\operatorname{id}_X \times x} X \times X \xrightarrow{m} X.$$

If $f : T \rightarrow \operatorname{Spec} k$ is a k -scheme, then on the T -points, t_x is simply multiplication by $x \circ f$ on the right. Similarly, we can define the left translation t'_x . Indeed, if $x \in X(T)$ for a k -scheme T , we can define $t_x, t'_x : X_T \rightarrow X_T$ similarly.

Abelian varieties are commutative as the name indicates, this result is based on the rigidity lemma.

Lemma 1.4 (Rigidity Lemma). *Let X, Y, Z be separated k -schemes of finite type that are geometrically integral. Suppose that X is proper. If $f : X \times Y \rightarrow Z$ is a morphism with the property that, for some $y \in Y(k)$, the fiber $X \times \{y\}$ is mapped to a point $z \in Z(k)$, then f factors through the projection $X \times Y \rightarrow Y$.*

Proof. See [1, Lemma 1.12]. □

Corollary 1.5. *Let $f : X \longrightarrow Y$ be a morphism of abelian varieties. Then f is the composition $t_{f(e_X)} \circ h$, where h is a homomorphism. In particular, the inverse morphism i is a homomorphism and thus X is commutative.*

Proof. Let $h = t_{i_Y \circ f(e_X)} \circ f$. It suffices to show that h is a homomorphism. By construction, we have $h(e_X) = e_Y$. Consider the composition morphism

$$g : X \times X \xrightarrow{(hom) \times (iom \circ (h \times h))} Y \times Y \xrightarrow{m} Y.$$

On T -points, this morphism is given by

$$(x, x') \mapsto h(xx')[h(x)h(x')]^{-1}.$$

We have $g|_{\{e_X\} \times X} = \{e_Y\} = g|_{X \times \{e_X\}}$. By the rigidity lemma, this implies that g factors both through the first and through the second projection $X \times X \longrightarrow X$. This means g is the constant map with value e_Y . This completes the proof. $\square$

Notation 1.6. From now on, we can simply use the additive notation for abelian varieties. Given two morphism $f, g : X \longrightarrow Y$ with Y a commutative group scheme, we define

$$f + g := m \circ (f, g), \quad \text{and} \quad -f := i \circ f = f \circ i.$$

It follow that $\text{Hom}(X, Y)$ is an abelian group.

2 Line bundles on Abelian varieties

Let X, Y be k -schemes with $x \in X$. If L is a line bundle on $X \times Y$, we write L_x for the restriction of L to $\{x\} \times Y$. Strictly speaking, L_x is pullback of L along $\text{Spec } k(x) \times Y \longrightarrow X \times Y$. We use the same notation for the analogous restrictions with respect to the other factors.

Theorem 2.1. *Let X, Y be proper k -schemes that are geometrically integral and let Z be a connected locally noetherian k -scheme. Consider points $x \in X, y \in Y$ and $z \in Z$. If L is a line bundle on $X \times Y \times Z$ such that L_x, L_y and L_z are trivial, then L is trivial.*

Proof. See [1, Theorem 2.5]. $\square$

Now we will state and prove the theorem of the cube. First, we introduce a notational convention. If X is an abelian variety and $I = \{i_1, \dots, i_r\} \subseteq \{1, 2, \dots, n\}$, then we write

$$p_I = p_{i_1 \dots i_r} : X^n \longrightarrow X$$

for the morphism sending $(x_1, \dots, x_n)$ to $x_{i_1} + \dots + x_{i_r}$. Thus for example p_i is the projection on the i -th factor and $p_{12} = p_1 + p_2$.

Theorem 2.2 (Theorem of the Cube). *Let L be a line bundle on X . The following line bundle is trivial on $X \times X \times X$:*

$$p_{123}^*L \otimes p_{12}^*L^{-1} \otimes p_{13}^*L^{-1} \otimes p_{23}^*L^{-1} \otimes p_1^*L \otimes p_2^*L \otimes p_3^*L.$$

Proof. The restriction of this bundle on $\{0\} \times X \times X \cong X \times X$ is

$$m^*L \otimes p_1^*L^{-1} \otimes p_2^*L^{-1} \otimes m^*L^{-1} \otimes \mathcal{O}_{X \times X} \otimes p_1^*L \otimes p_2^*L,$$

which is obviously trivial. Similarly for $X \times \{0\} \times X$ and $X \times X \times \{0\}$. By Theorem 2.1, the result follows. $\square$

Corollary 2.3. *Let X be an abelian variety and Y a scheme. For every triple $f, g, h : Y \rightarrow X$ of morphisms and for every line bundle L on X , the following line bundle is trivial:*

$$(f + g + h)^*L \otimes (f + g)^*L^{-1} \otimes (f + h)^*L^{-1} \otimes (g + h)^*L^{-1} \otimes f^*L \otimes g^*L \otimes h^*L.$$

Proof. This line bundle is the pullback of the trivial bundle in Theorem 2.2 along $(f, g, h) : Y \rightarrow X \times X \times X$, so is also trivial. $\square$

Theorem 2.4 (Theorem of the Square). *Let X be an abelian variety and let L be a line bundle on X . Then for all $x, y \in X(k)$, we have*

$$t_{x+y}^*L \otimes L \cong t_x^*L \otimes t_y^*L.$$

*In particular, the map $\varphi_L : X(k) \rightarrow \text{Pic}(X)$ given by $x \mapsto [t_x^*L \otimes L^{-1}]$ is a group homomorphism.*

Proof. The isomorphism is immediately from Corollary 2.3, by taking f the identity map on X and for g and h the constant maps with image x and y . In this case, we have

$$t_{x+y}^*L \otimes t_x^*L^{-1} \otimes t_y^*L^{-1} \otimes L \cong \mathcal{O}_X.$$

Therefore, a direct computation shows that φ_L is a group homomorphism. $\square$

The group homomorphism φ_L plays a very important role in the theory. Latter, I will introduce the Picard scheme $\text{Pic}_{X/k}$ and define the dual abelian variety X^t . We shall interpret φ_L as a homomorphism $X \rightarrow X^t$.

Recall that we have the multiplication-by- n map $[n] : X \rightarrow X$, which is realized as the multiplication-by- n on each abelian group $X(T)$ for any k -scheme T . The effect of $[n]$ on line bundle is described as follows.

Theorem 2.5. *For every line bundle L on an abelian variety X , we have*

$$[n]^*L \cong L^{n(n+1)/2} \otimes [-1]^*L^{n(n-1)/2}.$$

Proof. Set $f = [n], g = [1]$ and $h = [-1]$ and apply Corollary 2.3. It follows that

$$[n]^*L \otimes [n+1]^*L^{-1} \otimes [n-1]^*L^{-1} \otimes [n]^*L \otimes L \otimes [-1]^*L$$

is trivial and hence

$$[n+1]^*L \cong [n]^*L^2 \otimes [n-1]^*L \otimes L \otimes [-1]^*L.$$

The assertion follows by induction on n , starting from $n = -1, 0, 1$. $\square$

3 Quotients and Isogenies

First, we give a short introduction to the quotient of a scheme X/S by a group scheme G/S .

Definition 3.1. A group action of G on X is a morphism $\rho : G \times X \longrightarrow X$ such that for any S -scheme T , ρ induces a $G(T)$ -action on the set $X(T)$.

We would like to construct the "quotient" $G \backslash X$. It is a universal morphism of schemes $X \longrightarrow Y$ that is G -invariant.

Definition 3.2. Let X be a scheme with G -action. A morphism $f : X \longrightarrow Y$ is said to be G -invariant if the induced map $X(T) \longrightarrow Y(T)$ is $G(T)$ -invariant in the sense that $f(gx) = f(x)$.

We will skip the technical details and simply state that the quotient exists when X is an abelian variety and G is a closed subgroup scheme. The idea is to view X as an fppf sheaf valued in the category of abelian groups. We then construct the quotient sheaf and use its representability to obtain the quotient. For more details, one can view [1, Chapter IV].

Moreover, when we say that a sequence of group schemes is exact, we mean that the corresponding sequence of fppf sheaves is exact.

Proposition 3.3. *Let $f : X \longrightarrow Y$ be a homomorphism of abelian varieties. The following are equivalent:*

1. f is surjective and $\dim X = \dim Y$;
2. $\ker f$ is a finite group scheme and $\dim X = \dim Y$;
3. f is finite flat and surjective.

Proof. See [1, Proposition 5.2]. □

Definition 3.4. A homomorphism $f : X \longrightarrow Y$ of abelian varieties is said to be an isogeny if it satisfies the three equivalent conditions in Proposition 3.3. The degree of an isogeny is the degree of the function field extension $[k(X) : k(Y)]$.

If $f : X \longrightarrow Y$ is an isogeny, we have an isomorphism $Y \cong X/\ker f$, which indicates that $f_*\mathcal{O}_X$ is a locally free $\mathcal{O}_Y$ -module of finite rank. Since Y is connected, computing this rank at the generic point of Y and $0 \in Y$, we obtain

$$\deg f = \operatorname{rank}_{\mathcal{O}_Y}(f_*\mathcal{O}_X) = \operatorname{rank}(\ker(f)) =: \dim_k \Gamma(\ker f, \mathcal{O}_{\ker f}).$$

Proposition 3.5. *Let $f : X \longrightarrow Y$ be an isogeny. The following are equivalent:*

1. $k(X)/k(Y)$ is a separable extension;
2. f is an étale morphism;
3. $\ker f$ is an étale group scheme.

Proof. This is the first half of proposition 5.6 in [1]. $\square$

Definition 3.6. An isogeny $f : X \rightarrow Y$ is said to be separable if it satisfies the equivalent conditions in Proposition 3.5.

Fact 3.7. For any $n \neq 0$, $[n]$ is an isogeny. If $g = \dim X$, we have $\deg[n] = n^{2g}$. If $\text{char } k \nmid n$, $[n]$ is separable.

Corollary 3.8. If $\text{char } k \nmid n$, then $X[n](k_s) = X[n](\bar{k}) \cong (\mathbb{Z}/n\mathbb{Z})^{2g}$. Here $X[n]$ is the kernel of $[n]$.

Proof. We know $X[n]$ is a finite etale k -group scheme, so we have

$$X[n] = \bigsqcup_{x \in X[n]} \text{Spec } k(x),$$

where each $k(x)$ is a finite separable extension of k . Since $\text{rank ker } f = n^{2g}$, we know $X[n](k_s) = X[n](\bar{k})$ is an abelian group of order n^{2g} killed by n . Further, for any factor $d \mid n$, the subgroup killed by d is $X[d](k_s)$, which is of order d^{2g} . Therefore, we must have the required isomorphism. $\square$

Proposition 3.9. If $f : X \rightarrow Y$ is an isogeny of rank d , then there exists an isogeny $g : Y \rightarrow X$ such that $f \circ g = [d]_Y$ and $g \circ f = [d]_X$.

Proof. $\ker f$ is a finite group scheme of rank d , therefore is killed by $[d]$. This means $[d]$ is $\ker f$ -invariant. The universal property of the quotient $X \rightarrow Y \cong X/\ker f$ provides the required isogeny $g : Y \rightarrow X$. Now we already have $g \circ f = [d]_X$. The equality $f \circ g = [d]_Y$ follows from the surjectivity of f and $f \circ g \circ f = f \circ [d]_X = [d]_Y \circ f$. $\square$

Definition 3.10. If there is an isogeny $f : X \rightarrow Y$, we say X and Y are isogenous. This defines an equivalence relation.

We now discuss the p -power torsion $X[p^n]$ when $\text{char } k = p$. In this case, $X[p^n]$ is no longer etale, and there is additional structure in the corresponding group scheme.

Fact 3.11. Suppose $\text{char } k = p$. There is an integer $f(X)$ with $0 \leq f \leq g = \dim X$ such that $X[p^m](\bar{k}) \cong (\mathbb{Z}/p^m\mathbb{Z})^f$ for all $m \geq 0$. If X and Y are isogenous, then $f(X) = f(Y)$.

Definition 3.12. Such $f(X)$ is called the p -rank of X .

In fact, the p -rank $f(X)$ can be any natural number $\leq g = \dim X$.

4 Picard functor and dual abelian variety

Given a scheme X , recall that $\text{Pic}(X)$ is the group of isomorphism classes of line bundles on X . Now let X be an S -scheme. Consider the functor sending any S -scheme T to the abelian group $\text{Pic}(X_T)$. Equipped with the fppf topology, this functor is not even a sheaf, so we first sheafify it. We hope that the sheafified

functor $\text{Pic}_{X/S}$ is representable, however, this is not true in general. But at least for abelian varieties X/k , the answer is yes. And we have an explicit description of $\text{Pic}_{X/k}(T)$ for any k -scheme T :

$$\text{Pic}_{X/k}(T) \cong \text{Pic}(X_T)/\text{pr}_T^* \text{Pic}(T) \cong \{(L, \alpha) : L \in \text{Pic}(X_T), \alpha : \mathcal{O}_T \cong e_T^* L\}.$$

Definition 4.1. Taking the connected component of the neutral element of the Picard scheme, we define $X^t := \text{Pic}_{X/k}^0$. This is an abelian variety and is called the dual of X .

Now we will upgrade φ_L to a homomorphism $X \rightarrow X^t$ for line bundle L on X . It suffices to give the homomorphism $\varphi_L : X \rightarrow \text{Pic}_{X/k}$ since X is connected. For a while, we write $P := \text{Pic}_{X/k}$ for simplicity.

Definition 4.2. Any line bundle on $X \times P$ that corresponds to the identity map $\text{id}_P \in P(P)$ is called a Poincare bundle on X .

For any k -scheme T and line bundle L over X_T . L corresponds to an element f in $P(T)$. The representability of P tells us that L is the pullback of the Poincare bundle along $(\text{id}_X \times f)$. This is the universal property of the Poincare bundle.

Now we define φ_L to be the morphism in $P(X)$ that corresponds to the Mumford bundle $\Lambda(L) := m^* L \otimes p_1^* L^{-1} \otimes p_2^* L^{-1}$. It is the unique morphism such that

$$(\text{id}_X \times \varphi_L)^* \mathcal{P} \cong \Lambda(L).$$

For any k -scheme T , we compute the induced map of φ_L on T -valued points. Take $x \in X(T)$. We have the following commutative diagram

$$\begin{array}{ccc} P(T) = \{\text{line bundle on } X \times T\} / \cong & & \varphi_L(x) = (\text{id}_X \times x)^* \Lambda(L) \\ \uparrow & & \uparrow \\ P(X) = \{\text{line bundle on } X \times X\} / \cong & & \varphi_L = \Lambda(L) \\ \uparrow & & \uparrow \\ P(P) = \{\text{line bundle on } X \times P\} / \cong & & \text{id}_P = \mathcal{P} \end{array}$$

Therefore, $\varphi_L(x)$ corresponds to the isomorphism class

$$[(\text{id}_X \times x)^* \Lambda(L)] = [t_x^* L \otimes L^{-1} \otimes x^* L^{-1}] = [t_x^* L \otimes L^{-1}].$$

Fact 4.3. For every ample line bundle L on X , $\varphi_L : X \rightarrow X^t$ is an isogeny with kernel $K(L)$, which is the maximal closed subscheme of X such that $\Lambda(L)|_{X \times K(L)}$ is trivial in $P(K(L))$.

Definition 4.4. A polarization of an abelian variety X/k is an isogeny $\lambda : X \rightarrow X^t$ such that there exists a finite separable extension K/k and an ample line bundle L on X_K such that $\varphi_L = \lambda_K$. We say λ is a principal polarization if it is a polarization of degree 1.

References

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